Q1

A builder appoints three construction workers Akash, Sunil and Rakesh on one of his sites. They take 20, 30 and 60 days respectively to do a piece of work. How many days will it take Akash to complete the entire work if he is assisted by Sunil and Rakesh every third day?

Solution:

Total work done by Akash, Sunil and Rakesh in 1 day = {(1/20) + (1/30) + (1/60)} = 1/10

Work done along by Akash in 2 days = (1/20) × 2 = 1/10

Work Done in 3 days (1 day of all three together + 2 days of Akash's work) = (1/10) + (1/10) = 1/5

So, work done in 3 days = 1/5

Time taken to complete the work = 5×3 = 15 days

Q2:

To complete a piece of work, Samir takes 6 days and Tanvir takes 8 days alone respectively. Samir and Tanvir took Rs.2400 to do this work. When Amir joined them, the work was done in 3 days. What amount was paid to Amir?

Solution:

Total work done by Samir and Tanvir = {(1/6) + (1/8)} = 7/24

Work done by Amir in 1 day = (1/3) – (7/24) = 1/24

Amount distributed between each of them = (1/6) : (1/8) : (1/24) = 4:3:1

Amount paid to Amir = (1/24) × 3 × 2400 = Rs.300

Q3

Time taken by A to finish a piece of work is twice the time taken B and thrice the time taken by C. If all three of them work together, it takes them 2 days to complete the entire work. How much work was done by B alone?

Solution:

Time taken by A = x days

Time taken by B = x/2 days

Time Taken by C = x/3 days

$\Rightarrow \{(1/x) + (2/x) + (3/x) = 1/2$

$\Rightarrow 6/x = 1/2$

$\Rightarrow x = 12$

Time taken by B = x/2 = 12/2 = 6 days